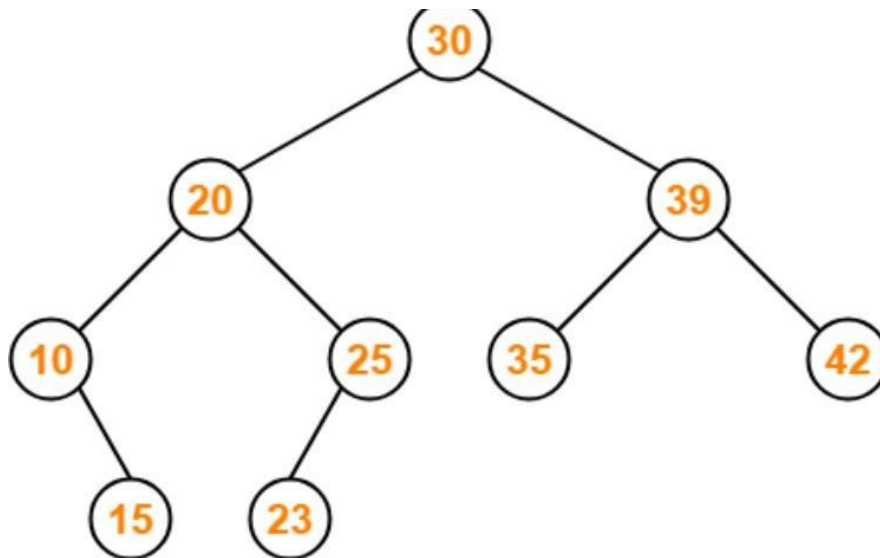


CSA0309 – Data Structures (Slot B)

CO3 - AT2 - Tree Traversal Challenges

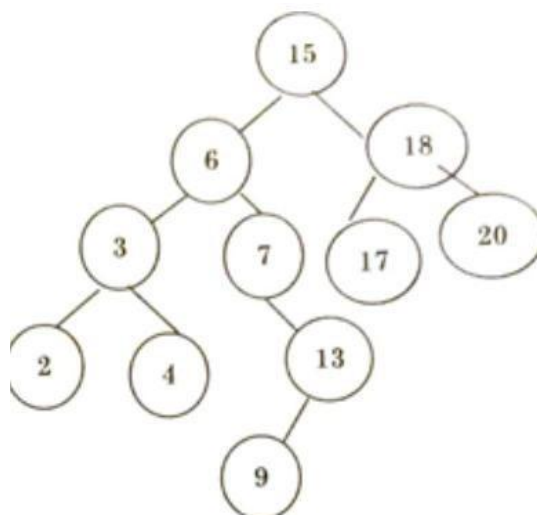
Question No:1

Write the traversal sequences for following trees



Question No: 2

Write the Inorder, Preorder & Postorder traversal sequences for following trees



Question No: 3

Consider the following Inorder and Preorder traversal of a tree

Inorder: D B E F A G H C

Preorder: A B D E F C G H

What is the Postorder traversal of the same?

1. D F E B H C G A

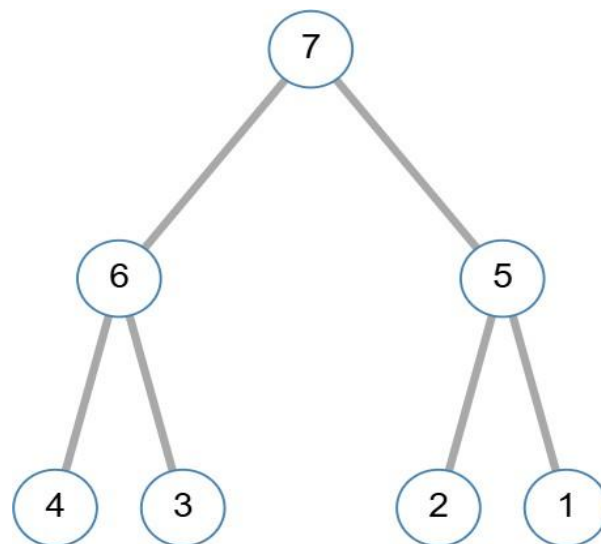
2. D F E B G H C A

3. D F E B H G C A

4. D F E H B G C A

Question No: 4

Find Inorder, Preorder and Postorder



Question No: 5

Assume the digits 7, 5, 1, 8, 3, 6, 0, 9, 4, 2 are added in that sequence into a binary search tree that is initially empty. The binary search tree follows the standard natural number ordering. What is the resulting tree's traversal sequence?

1)

[← Back to Home](#)

Tree Traversals

Input

Input is level-by-level order.

30,20,39,10,25,35,42,15,23

Enter numbers separated by commas. Use 'null' for empty nodes.

Traversal Order

Pre

In

Post

Generate Default

Visualize

Start Simulation

30 20 10 15 23 25 39 35 42

```
graph TD; 30((30)) --> 20((20)); 30 --> 39((39)); 20 --> 10((10)); 20 --> 25((25)); 10 --> 15((15)); 10 --> 23((23)); 39 --> 35((35)); 39 --> 42((42)); style 42 fill:#f96
```

2) Preorder

[← Back to Home](#)

Tree Traversals

Input

Input is level-by-level order.

15,6,3,7,13,17,20,2,4,9

Enter numbers separated by commas. Use 'null' for empty nodes.

Traversal Order

Pre

In

Post

Generate Default

Visualize

Start Simulation

15 6 7 2 4 13 9 3 17 20

```
graph TD; 15((15)) --> 6((6)); 15 --> 3((3)); 6 --> 7((7)); 6 --> 13((13)); 7 --> 2((2)); 7 --> 4((4)); 13 --> 9((9)); 3 --> 17((17)); 3 --> 20((20)); style 20 fill:#f96
```

Inorder

[← Back to Home](#)

Tree Traversals

15 6 7 2 4 13 9 3 17 20

Input

Input is level-by-level order.

15,6,3,7,13,17,20,2,4,9

Enter numbers separated by commas. Use 'null' for empty nodes.

Traversal Order

Pre In Post

Generate Default

Visualize

Start Simulation

Postorder

[← Back to Home](#)

Tree Traversals

2 4 7 9 13 6 17 20 3 15

Input

Input is level-by-level order.

15,6,3,7,13,17,20,2,4,9

Enter numbers separated by commas. Use 'null' for empty nodes.

Traversal Order

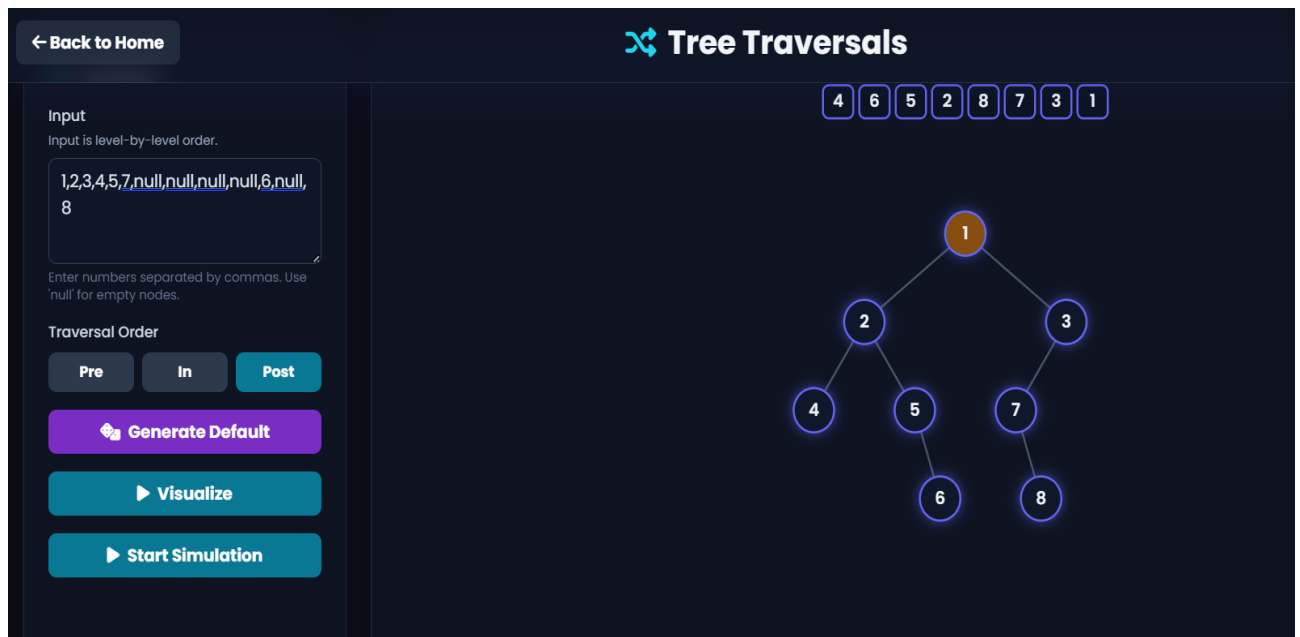
Pre In Post

Generate Default

Visualize

Start Simulation

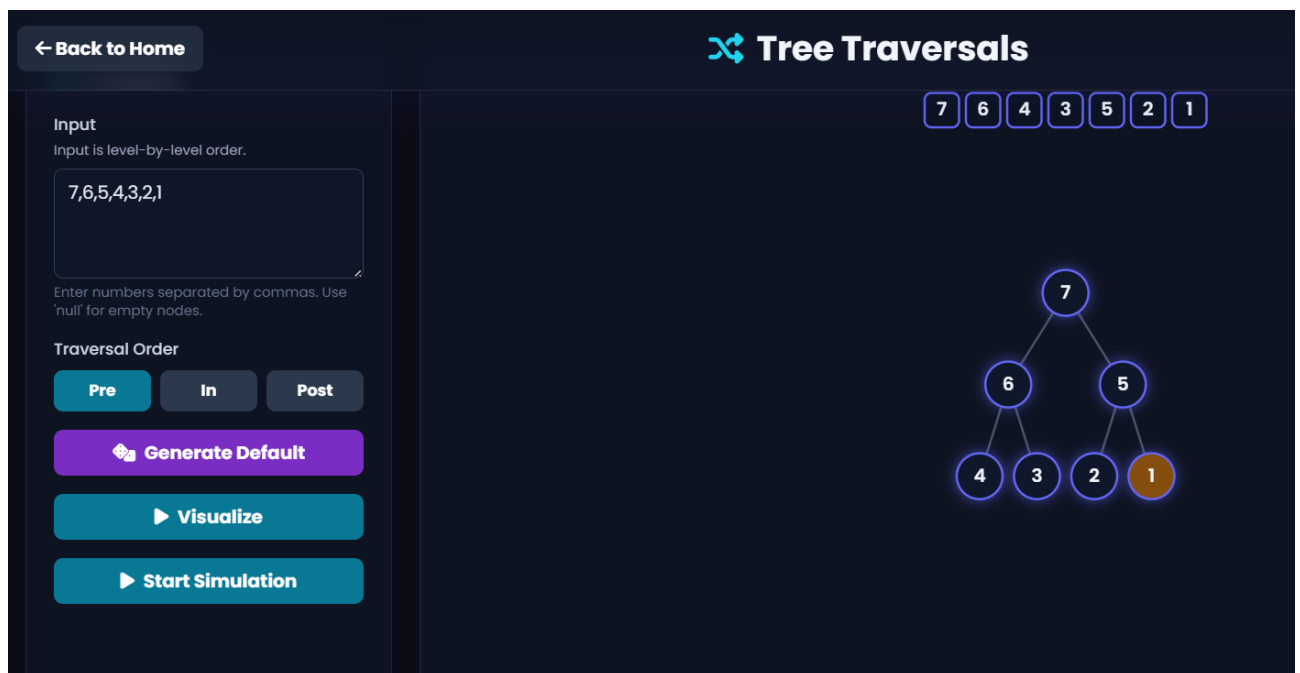
3)



Option (3)

4)

Preorder



Inorder

[← Back to Home](#)

Tree Traversals

4 6 3 7 2 5 1

Input

Input is level-by-level order.

7,6,5,4,3,2,1

Enter numbers separated by commas. Use 'null' for empty nodes.

Traversal Order

Pre In Post

Generate Default

Visualize

Start Simulation

Postorder

[← Back to Home](#)

Tree Traversals

4 3 6 2 1 5 7

Input

Input is level-by-level order.

7,6,5,4,3,2,1

Enter numbers separated by commas. Use 'null' for empty nodes.

Traversal Order

Pre In Post

Generate Default

Visualize

Start Simulation

5)

Enter a number:



INSERT

DELETE

CLEAR

